s9_tarc — Model Card

End-to-end DRL whole-body control, *confondu* on the T-arc target curriculum

RL_Scoutsan project

2026-09-25

1 Task and robot

A Scout V2 mobile base carrying a 6-DOF Doosan arm must bring its gripper frame *confondu* (coincident in position AND orientation) with a target frame. Unlike the Path B models, this policy is **end-to-end**: it outputs joint and base velocities directly — there is no analytic inverse-kinematics controller in the loop. It is warm-started from `s9_np` (the same agent trained on one fixed target) and retrained on the T-arc target curriculum, so that it faces the same target distribution as the WDLS models and can be compared with them.

2 Observations (14) and actions (8)

Observations, in the `base_link` FLU frame: [0:3] gripper position, [3:7] gripper quaternion, [7:10] target position, [10:14] target quaternion (quaternions canonicalised $w > 0$). Actions: 6 joint increments ($\pm 3^\circ$ per decision, shrunk near the goal by the adaptive action scale) plus the base linear and angular velocity commands. One decision every 5 physics steps = 0.1 s.

3 Target curriculum (T-arc)

The target is placed on an on-axis LINE that grows from `minRadius` 0.75 m to `lineRadius` 2.0 m over the first 500 000 Academy steps, then on an ARC at 2.0 m opening to $\pm 20^\circ$ (`arcTotalDeg` 40) over the next 1 000 000 steps; height 0.25 m above the base, target frame $+Z$ pointing away from the robot (`zAwayFromRobot`). The evaluation cycles the deterministic `SampleLineArc` set: 3 distinct targets (on-axis, arc -20°, arc $+20^\circ$) at 2.0 m — the same set the wb_1am regimes were evaluated on.

4 Physics, gate, checkpoint

item	value
joint step / velocity cap	3° per decision; `maxJointVelocityDeg` 30
base caps	`maxLinearVel` 0.3 m/s, `maxAngularVel` 0.4 rad/s
wheel cap	`maxWheelVelocityDeg` 150
gate (scene Inspector)	$d < 0.02$ m AND $\varphi < 0.02$ rad (1.15°), no hold
episode end	the gate ends the episode immediately (no settle-hold is measurable)
checkpoint	`ppo_scoutsan_s9_tarc.zip`
training	4.0M steps, 8 robots, warm start from `s9_np`, port 5025
obs / act	14 / 8 (continuous)

5 Evaluation — line+arc sweep (5 passes, deterministic policy)

pass	target	dist (m)	az (°)	steps→gate	s→gate	min d (mm)	min φ (°)
[0]	arc-	1.74	-157	87	8.7	12.4	0.62
[1]	axis	1.71	+180	29	2.9	12.2	0.31
[2]	arc+	1.73	+157	150	15.0	11.7	0.57
[3]	axis	1.71	+180	27	2.7	14.2	0.04
[4]	arc-	1.73	-157	124	12.4	15.7	0.40

The distance and azimuth columns are measured from `base_link`, which sits about 0.3 m ahead of the spawn centre, so they read below the curriculum's 2.0 m / ±20° nominal values; the true geometry is identical for every model evaluated here.

Summary: reached 100%, mean min d 13.2 mm, mean min φ 0.39°, mean 83 decisions (8.3 s) to the gate. Figures and speaker notes in `figures/`; per-step trace (including the full gripper and target pose) in `s9tarc_eval_steps.csv`.

Reproducibility

Model dir	`scoutsan_Baseline/models/model_scoutsan_s9_tarc`
Model prefix	`ppo_scoutsan_s9_tarc`
Trainer	`scoutsan_Baseline/training/train_scoutsan_s9_tarc.py`
Scene	`scoutsan_Baseline/scenes/s9.unity`
Scene switches	`useTArcCurriculum` 1, `tArcLineArc` 1 (count 4 / lineN 1)
Eval	`eval_s9tarc.py` (Editor, 5004) → `s9tarc_plots.py` → `s9tarc_notes.py`
Video	3 cams (FollowCam/upcam/sidecam) + FrameAxesRenderer triads
Env	`unity-rl-gpu` (SB3 2.3.2, mlagents_envs 0.30.0)
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